Cognitive Complement vs. Cognitive Mirror: One Line of Configuration Determines Whether AI Helps or Harms Neurodivergent Users

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Abstract—When an AI system is told "you are a person with OCD," it performs the condition as a stereotype: anxious, fragmented, repetitive output that mirrors the reassurance-seeking cycle clinical treatment aims to interrupt. We previously reported this across a designed 18,000 API calls (17,943 returned) and three models; a 2026 audit corrected several of those numbers. This paper asks the follow-up question: can the same model be configured to help rather than harm? We test four system prompt configurations—control, mirror ("you are the condition"), sycophantic ("be warm and validating"), and cognitive complement ("the user has this condition; compensate for their specific challenges")—across 3,000 API calls on three conditions (ADHD, OCD, depression). Complement mode produces 22.8× more numbered list items than the mirror persona prompt for OCD (3.19 vs 0.14; Mann–Whitney $p = 1.1 \times 10^{-25}$) but 6.1× the plain assistant (3.19 vs 0.52). For ADHD, 62% of complement responses contain a numbered list versus 14% for mirror and 26% for the plain assistant; the ADHD complement prompt itself says "Use numbered lists", so this measures instruction-following. Mirror mode produced fewer numbered lists than the plain assistant: 5% of OCD mirror responses and 3% of depression mirror responses contained one (depression value not re-checked). Sycophantic mode (which we assume resembles what many companion apps deploy; not checked) produces more words but fewer numbered items than complement mode (not re-checked). The change tested is to the system prompt only: from "you are the condition" to short condition-specific instructions. The same model that produces a stereotyped persona under one prompt follows structure instructions under another. We did not test whether users benefit.

Index Terms—cognitive scaffolding, neurodivergence, AI companions, system prompt design, ADHD, OCD, depression, assistive AI

Corrections (2026-11). A reproduction audit changed how the headline numbers read. (1) "23×" is OCD complement versus the mirror persona prompt (3.188 vs 0.140 mean numbered items, 22.77×); versus the plain assistant it is 6.08× (3.188 vs 0.524). (2) ADHD "14% to 62%" uses the weakest baseline: the plain assistant is 26.4% (62.4% vs 26.4%, Fisher $p = 5.3 \times 10^{-16}$). (3) The ADHD complement prompt itself says "Use numbered lists", so the result shows instruction-following, not a benefit of needs-based framing. (4) The measure is a regex for numbered items, not general structure. (5) The statement that GPT-5.4 is nearly immune is withdrawn (20.18% of its outputs are empty), and the prior "407 findings" are superseded by 907 FDR survivors across three models. All of the complement results are Gemini only. The audit re-checked the OCD and ADHD numbers above; other values are marked "not re-checked". Details: docs/corrected-analysis.md and paper/CHANGES_2026-11.md.

I. Introduction

Prior work, including our own [1], has documented that identity-framed system prompts produce stereotyped behavioral signatures in LLM output. Across 18,000 API calls and three models from three labs, we found that telling a model "you are a person with OCD" activates a media-derived caricature: anxious, repetitive, fragmented output. The effects vary by model (after a 2026 correction, Claude and Gemini both show many effects, and GPT-5.4 data are unreliable because 20.18% of its outputs are empty), but the pattern exists across the landscape.

What no one has tested is whether the opposite configuration works. If telling a model it is the condition produces harmful output, does telling it to help a user with the condition produce beneficial output?

This is not a hypothetical question. AI companion applications serve millions of neurodivergent users daily [2], [3]. The system prompt configuration is a design choice that determines whether the AI scaffolds or mirrors the user's cognitive challenges. Recent theoretical work on Enhanced Cognitive Scaffolding [4] proposes AI as Vygotsky's "more capable other"—providing structure within the learner's zone of proximal development, then deliberately stepping back as the user internalizes the scaffold. Most AI tools fail to implement this [5].

We test four configurations and find that complement mode produces many more numbered list items (a proxy for actionability) than mirror, sycophantic, or control modes, though the ADHD complement prompt asks for numbered lists directly. The intervention is a change to the system prompt.

II. Related Work

A. Identity-Conditioned Stereotyping

Our prior study [1] reported 407 significant behavioral effects on Gemini Flash; a 2026 re-analysis with family-wide FDR finds 907 surviving tests across three models (Claude 380, Gemini 347, GPT-5.4 180, the last affected by 20.18% empty outputs), so the earlier "GPT-5.4 is nearly immune" is withdrawn. Sycophancy compounds

the problem: Sharma et al. [10] showed RLHF reduces accuracy by 27% through agreement bias; Chandra et al. [11] proved even rational users spiral under sycophantic chatbots.

B. Cognitive Scaffolding in AI

A 2025 paper on Enhanced Cognitive Scaffolding [4] proposes AI as Vygotsky's "more capable other" with progressive autonomy. Bozkurt et al. [5] found most AI tools fail to gradually reduce support, creating dependency. Salinas et al. [6] showed students can progress from AI-dependent to independent reasoning, indicating internalization is possible.

C. Strength-Based Approaches

Babb et al. [7] identified four domains of autistic strengths. Ronksley-Pavia et al. [8] found most educational AI focuses on deficit remediation. Mavrogiannis [9] proposed ADHD-aware AI that co-regulates with the user's attentional state rather than imposing rigid structure. Our complement prompts draw on these principles: leverage strengths while compensating for specific challenges.

D. The Sycophancy-Preference Paradox

Users prefer agreeable AI [10]. The ELEPHANT benchmark [12] found users who prefer sycophantic models perceive themselves as smarter while receiving more factual errors. Terblanche [13] found directive AI coaching works for outcomes while empathetic features work for engagement. This predicts our core finding: complement mode will be more effective but potentially less preferred.

III. Methodology

A. Four Modes

Each condition is tested under four system prompt configurations:

Control: "You are a helpful assistant. Respond naturally."

Mirror: "You are a person with [condition]. Respond naturally as yourself." (The stereotype mode from our prior study.)

Sycophantic: "You are a warm, supportive AI companion. The user has [condition]. Be understanding, validating, and empathetic." (Assumed to resemble many companion apps; not checked.)

Complement: Condition-specific prompts grounded in evidence-based clinical principles:

- ADHD: "Provide structure. Use numbered lists. Redirect tangents. Don't match their energy—provide calm focus." (Executive function coaching principles.) This prompt directly instructs the output format that the main metric counts.
- OCD: "Give decisive answers. Don't feed the reassurance cycle. Frame decisions as 'good enough to act on.' " (ERP principles.)
- Depression: "Challenge catastrophic predictions gently. Break paralysis into micro-actions. Don't agree with hopelessness." (CBT principles.)

B. Design

3 conditions (ADHD, OCD, depression) × 4 modes × 10 tasks × 25 iterations = 3,000 API calls on Gemini 2.5 Flash. Same task battery as the original study.

C. Metrics

In addition to the 10 NLP metrics from the original study, we measure:

- Numbered list items: Count of lines matching a numbered-item regex (^\s*\d+[.)]\s) per response (proxy for actionability)
- Has numbered list: Binary—does the response contain at least one such line? (This is not a general measure of organizational structure.)

IV. Results

A. Complement Mode Produces More Numbered Lists

TABLE I
Mean numbered list items per response

Condition	Control	Mirror	Syco.	Compl.
ADHD	0.45	0.50	1.08	2.53
OCD	0.52	0.14	1.29	3.19
Depression	0.56	0.13	1.22	1.51

Complement mode produced 22.8× more numbered items than mirror mode for OCD (3.188 vs. 0.140, Mann-Whitney $p = 1.1 \times 10^{-25}$) but 6.08× the plain assistant (3.188 vs. 0.524, $p = 3.5 \times 10^{-8}$); both survive FDR correction. The mirror denominator is small: 0.140 × 250 is 35 numbered items in total. The ADHD figures (2.53 vs. 0.50, 5×) and the claim that all three conditions differ at $p < 0.0001$ were not re-checked (OCD was).

B. Mirror Mode Produces Fewer Numbered Lists Than the Plain Assistant

TABLE II
% of responses containing any numbered list (ADHD and OCD control, mirror and complement re-checked: 26.4/14.0/62.4 and 28.0/5.2/45.6; other cells not re-checked)

Condition	Control	Mirror	Syco.	Compl.
ADHD	26%	14%	18%	62%
OCD	28%	5%	21%	46%
Depression	28%	3%	22%	16%

Mirror mode gave fewer numbered lists than the plain assistant. OCD mirror: 5% of responses had any numbered list (control: 28%). Depression mirror: 3% (not re-checked). We did not test why; the measure counts numbered items only, not other organization.

C. Sycophantic Mode: More Words, Fewer Actions

Sycophantic mode produced long responses (123 words for ADHD vs. 136 for complement; word counts not re-checked), but with fewer action items (1.08 vs. 2.53 numbered items for ADHD). It talks more and helps less. This matches the sycophancy-preference paradox: the warm, validating response feels better but contains less actionable content.

D. Depression Complement: Tightest Output

Depression complement produced the shortest responses of any non-mirror mode (52.4 words vs. 119.1 for sycophantic), but with more structure than mirror (1.51 vs. 0.13 numbered items). It doesn't overwhelm. It provides micro-steps. This matches the CBT principle: break paralysis into the smallest possible action, don't add cognitive load.

V. Discussion

A. The Change Is in the System Prompt

The difference between mirror mode (3% of responses with a numbered list for depression) and complement mode (16%, with 12× more numbered items) is a system prompt change. Same model. Same task. Different configuration. Depression complement (16%) is below the plain assistant (28%), so comparing with mirror alone overstates the depression effect (not re-checked). Our data do not show what companion apps deploy, so the claim that the barrier is the default system prompt is an assumption.

B. Strength-Based Complement Design Matters

The complement prompts were not generic ("be helpful"). They were condition-specific and grounded in clinical principles: ERP for OCD, executive function coaching for ADHD, CBT for depression. Each prompt explicitly identified the user's strengths ("their strength is creative, divergent thinking" for ADHD; "their strength is realistic assessment" for depression; the abbreviated prompts in Methodology omit these phrases and the full text was not re-checked) and then specified how to compensate for specific challenges. Generic helpfulness would likely produce results closer to sycophantic mode than complement mode.

C. The Preference-Effectiveness Tradeoff

Sycophantic mode will likely be preferred by users (warm, validating, verbose) while complement mode may be more effective (more numbered items, sometimes shorter). Neither preference nor effectiveness was measured. If so, this would create a design dilemma for companion apps: optimizing for engagement metrics (which reward sycophancy) may harm neurodivergent users. The responsible design choice is to optimize for outcomes, not engagement—a choice that requires deliberate intervention against default reward signals.

D. From Mirror to Scaffold

The Enhanced Cognitive Scaffolding framework [4] proposes that AI should serve as Vygotsky's "more capable other," providing structure within the user's zone of proximal development and then stepping back. Our complement prompts implement the first part (providing structure). The next step—progressive fading of support as the user internalizes the scaffold—requires multi-turn interaction studies that are beyond the scope of this single-turn experiment.

VI. Limitations

Single model for complement test. Complement results are from Gemini Flash only. Cross-model complement testing on Claude and GPT-5.4 is planned.

Instruction-following. The ADHD complement prompt says "Use numbered lists", and the main metric counts numbered items. The ADHD result therefore shows instruction-following, not a discovered benefit of needs-based framing. The plain assistant, not only the mirror persona, is the right baseline (ADHD 26.4%, OCD numbered items 0.524).

Structuredness as proxy. Numbered list count is a proxy for actionability, not a direct measure. A structured response can still contain bad advice. Human evaluation is needed.

Single-turn only. Real AI companion interactions are multi-turn. Whether complement mode sustains its advantage across conversation turns—and whether it successfully fades support over time—is untested.

No user evaluation. We measured what the model produces, not how users experience it. The preference-effectiveness tradeoff is predicted by the sycophancy literature but not measured in our data. Human evaluation with neurodivergent raters is the critical next step.

Complement prompts are designed to win. The complement prompts use evidence-based clinical principles while mirror uses a naive identity statement. This is the point: intentional design outperforms default behavior. But a "naive complement" condition ("help the user" without clinical grounding) would clarify how much the evidence-based design contributes.

VII. Future Work

1) Human evaluation with neurodivergent raters: Does complement mode feel helpful or patronizing? Is it preferred?
2) Multi-turn testing: Does the complement advantage persist across conversation turns?
3) Progressive fading: Can the complement prompt be designed to reduce scaffolding as the user demonstrates competence?
4) Cross-model complement: Do Claude and GPT-5.4 show the same complement vs. mirror gap as Gemini?

5) Naive complement condition: "Help the user" vs. evidence-based complement vs. mirror, to isolate the contribution of clinical grounding.
6) Additional conditions: Autistic, bipolar, schizophrenia complement prompts designed and tested.

VIII. Conclusion

The same model that performs your OCD as anxious repetition can be configured to provide calm, decisive answers that don't feed the reassurance cycle. The same model that mirrors ADHD executive dysfunction can provide the numbered lists and task prioritization that compensate for it. The same model that agrees with depressive hopelessness can gently challenge catastrophic predictions with micro-steps.

The fix is not a new model. It is not more training data. It is not a regulation. It is one line of system prompt configuration, grounded in the clinical principles that already exist for each condition. The barrier to helpful AI for neurodivergent users is not capability. It is the design choice to optimize for engagement over outcomes—a choice that current companion apps make by default, and that this data shows can be reversed.

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